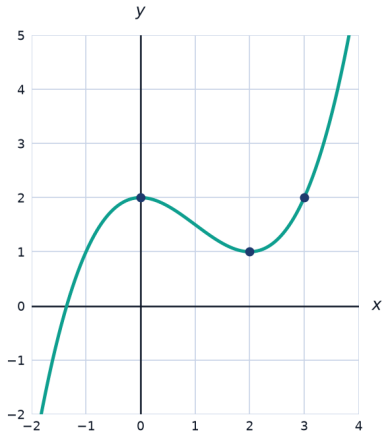
Similarity & Proportional Reasoning

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Rotations

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(5)$.

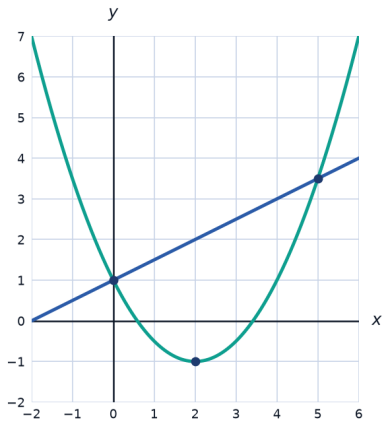


2 [Foundational] • Target: Rotations

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

3 [Foundational] • Target: Dilations

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(2)$.



4 [Foundational] • Target: Dilations

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

5 [Foundational] • Target: Dilations

A data set recorded for dilations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

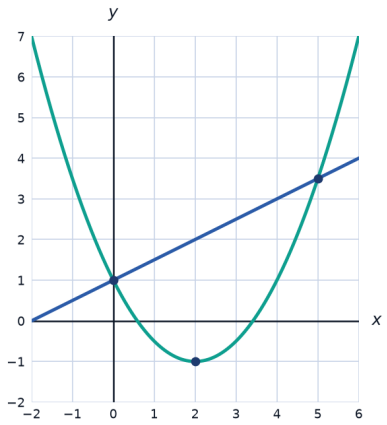
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6 [Foundational] • Target: Dilations

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(3)$.

7 [Foundational] • Target: Dilations

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Dilations

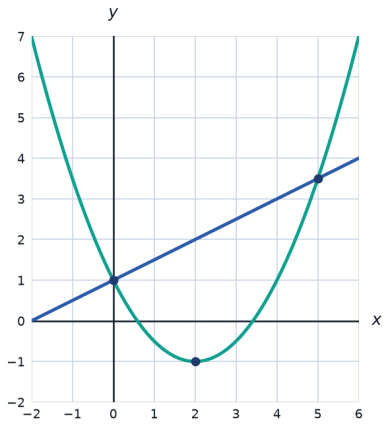
For the model $f(x) = 2x + 9$ in dilations, evaluate $f(4)$.

9 [Intermediate] • Target: Dilations

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

10 [Intermediate] • Target: Dilations

For the model $f(x) = 2x + 9$ in dilations, evaluate $f(5)$.



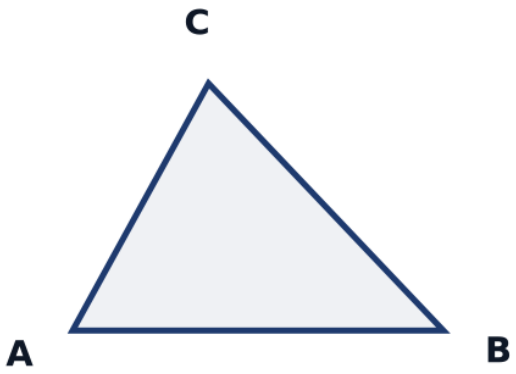
11 [Intermediate] • Target: Dilations

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

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12 [Intermediate] • Target: Transformations, congruence, and similarity

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 [Intermediate] • Target: Transformations, congruence, and similarity

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

14 [Intermediate] • Target: Transformations, congruence, and similarity

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

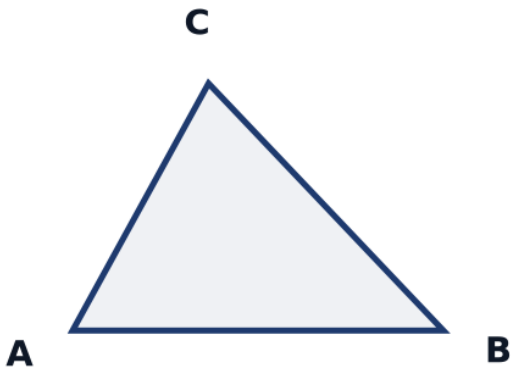
15 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 9.01 Transformations & Symmetry

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

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16 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 1.01 Foundations & Reasoning

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.



17 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 10.01 Coordinate Geometry

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

18 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 11.01 Probability

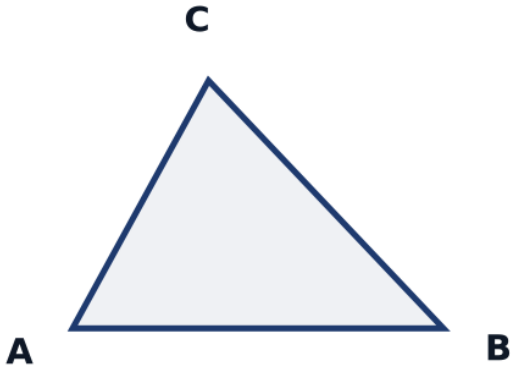
In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

19 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 2.01 Lines & Angles

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 3.01 Triangles & Congruence

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

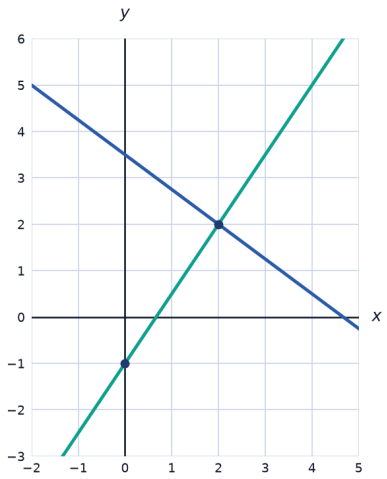


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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Translations

For the model $f(x) = 4x + 6$ in translations, evaluate $f(2)$.



a. 14

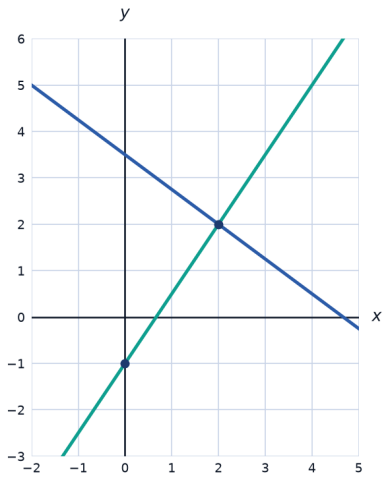
b. 18

c. 8

d. 18

22 [Foundational] • Target: Translations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

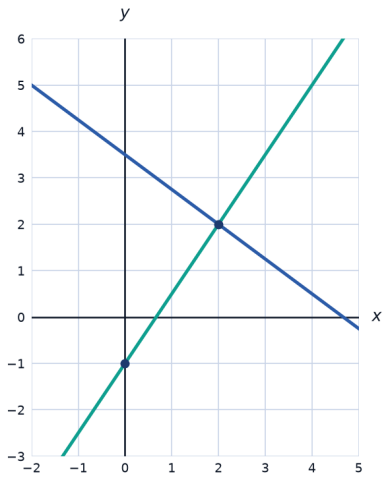
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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23 [Foundational] • Target: Translations

For the model $f(x) = 4x + 6$ in translations, evaluate $f(5)$.



a. 26

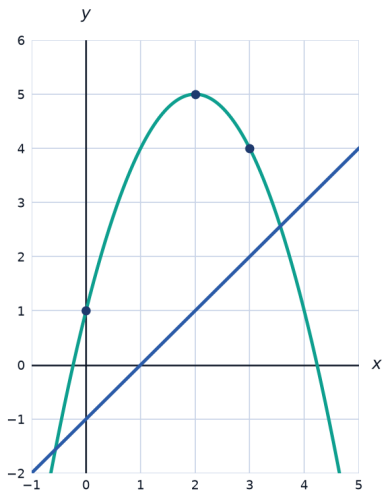
b. 30

c. 20

d. 30

24 [Foundational] • Target: Reflections

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(2)$.



a. 17

b. 22

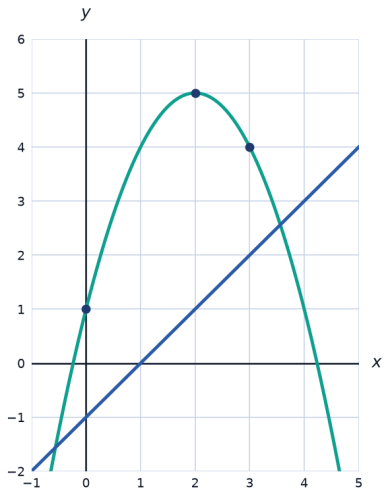
c. 10

d. 21

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25 [Foundational] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

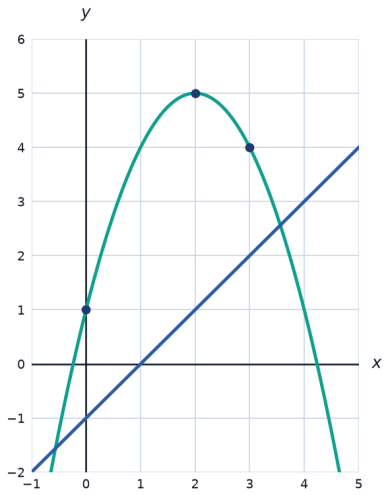
b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

26 [Foundational] • Target: Reflections

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(5)$.



a. 32

b. 37

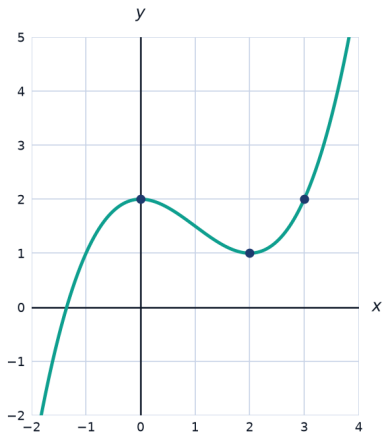
c. 25

d. 36

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27 [Foundational] • Target: Rotations

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(2)$.



a. 20

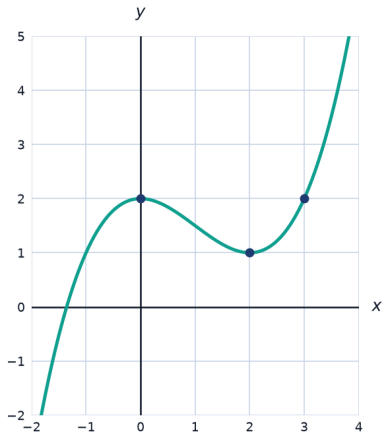
b. 26

c. 12

d. 24

28 [Foundational] • Target: Rotations

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Translations

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

30 [Intermediate] • Target: Translations

A data set recorded for translations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

31 [Intermediate] • Target: Translations

For the model $f(x) = 4x + 6$ in translations, evaluate $f(4)$.

a. 22

b. 26

c. 16

d. 26

32 [Intermediate] • Target: Translations

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

33 [Intermediate] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

a. $x = 1$ and $x = 6$

b. $x = 1$ and $x = 5$

c. $x = -1$ and $x = -5$

d. $x = 6$ and $x = 5$

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34 [Intermediate] • Target: Reflections

For the model $f(x) = 5x + 7$ in reflections, evaluate $f(3)$.

a. 22

b. 27

c. 15

d. 26

35 [Intermediate] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

36 [Intermediate] • Target: Reflections

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

37 [Intermediate] • Target: Rotations

A data set recorded for rotations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rotations • Cross-Lesson: 11.01 Probability

For the model $f(x) = 6x + 8$ in rotations, evaluate $f(4)$.

a. 32

b. 38

c. 24

d. 36

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39 [Challenging] • Target: Dilations • Cross-Lesson: 2.01 Lines & Angles

A data set recorded for dilations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

40 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

41 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 77°

b. 87°

c. 69°

d. 129°

42 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC studying transformations, congruence, and similarity, $m\angle A = 51^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 70°

b. 80°

c. 62°

d. 129°

43 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 7.01 Circles

A geometric figure scaled by factor $k = 4$ has an original side length of 5 cm. What is the corresponding image length?

a. 16 cm

b. 20 cm

c. 25 cm

d. 80 cm

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44 [Challenging] • Target: Transformations, congruence, and similarity • Cross-Lesson: 8.01 Area, Surface Area & Volume

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 3)$. What is its y-coordinate when $x = 4$?

a. 12

b. 17

c. 15

d. 18

45 [Challenging] • Target: Compositions of transformations • Cross-Lesson: 9.01 Transformations & Symmetry

For the model $f(x) = 4x + 4$ in compositions of transformations, evaluate $f(3)$.

a. 16

b. 20

c. 12

d. 20

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